Database Engineering (CS60032)

Assignment 1. R-Tree (Due Date: January 31, 2026, 11:59 PM)

Total Marks: 100 (30 Marks for R-Tree Creation program, 30 Marks for Nearest Neighbor Query program, 20 Marks for Data Generation program, 20 Marks for Results)

Write programs in any language to create and query R-Trees. You need to consider only Insert into R Trees and Nearest Neighbor query.

Three separate programs are to be written: One for insert, one for search and one for data creation. The Insert program will save the created R-Tree in a file. The Data Creation program will create and store test data in a file. The Search program will perform nearest neighbor search.

Test your code with synthetic data. Generate and Store test data using the Data Creation program in a file (to be used repeatedly). Format will be

d1min d1max d2min d2max d3min d3max … dnmin dnmax
d1min d1max d2min d2max d3min d3max … dnmin dnmax
d1min d1max d2min d2max d3min d3max … dnmin dnmax
..
N such n dimensional rectangles, one in each line.

Consider each dimension (both min and max) to have only randomly generated integer values between 0 and 20 (for each min, max pair, ensure that max is >= min). Take M=10 and m=4. N=10,000. Vary n as 2, 4, 8, 16, 32.

Overview of steps:

Step 1: Create dataset for n=2. Save it in a file.

Step 2: Create R-Tree for the dataset by reading the data stored in the file. Use Insert algorithm with quadratic split for handling node splitting. Save the R-Tree in a file for repeating search.

Step 3: Choose a random n-dimensional data point with value for each dimension in the integer range 0 to 20 as a query point.

Step 4: Implement Nearest Neighbor Search algorithm and query the R-tree (reading it from the file storing the R tree) with the input query point. While querying, note the time taken and the number of nodes visited. Save the data for reporting results. Note: For simplicity, consider that the underlying geometries are the same as the corresponding rectangles in the leaf node. Consider MINMAXDIST for determining the search order. Apply appropriate pruning strategies as discussed in class.

Step 5: Repeat Steps 3 to 4 twenty times and compute the average time (T) and average number of nodes visited (V). Convert V to the nearest integer.

Step 6: Repeat Steps 1 to 5 for n= 4, 8, 16, 32

Report your results in a table as follows:

Srl. No.	Number of Dimensions (n)	Average Time taken (msec) (T)	Average no. of Nodes visited (V)
1	2		
2	4		
3	8		
4	16		
5	32		

Submit your three separate programs and a pdf file containing the result in the above format through Moodle. **Note, you will have to give a demo of your working code on a suitable date later.**